

T.Y., B.Tech., CCE, Semester V Mini- Project

July- November, 2026

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Domain & Problem statement

Domain: A Hybrid Explainable AI Framework for FinTech Security Incident Investigation

Problem statement: Modern FinTech Security Operations Centers (SOCs) already employ AI-driven tools for threat detection and alert prioritization. However, security analysts still spend significant time correlating evidence across multiple security platforms, validating AI-generated insights, and preparing investigation reports for regulatory compliance. Existing solutions are often vendor-specific, rely on expensive cloud-based Large Language Models (LLMs), or provide limited explainability, making it difficult to justify security decisions in highly regulated financial environments. Therefore, there is a need for a **hybrid, explainable, and cost-effective AI framework** that integrates rule-based detection, anomaly analysis, and lightweight language models to automate evidence collection, incident investigation, and explainable reporting while keeping human analysts in the decision-making process.

Objectives

- **Design a Hybrid Investigation Framework:** Develop a hybrid framework that combines rule-based detection, anomaly detection, graph-based event correlation, and lightweight language models for FinTech security incident investigation.
- **Improve Explainability of AI Decisions:** Generate transparent investigation workflows, evidence trails, and explainable incident reports to support analyst decision-making and regulatory compliance.
- **Develop a Lightweight and Cost-Effective Solution:** Evaluate the use of small language models (SLMs) and hybrid reasoning techniques to reduce computational cost while maintaining investigation quality.
- **Evaluate Investigation Performance:** Compare the proposed framework with conventional investigation workflows using metrics such as investigation time, query accuracy, explainability, resource utilization, and report quality.

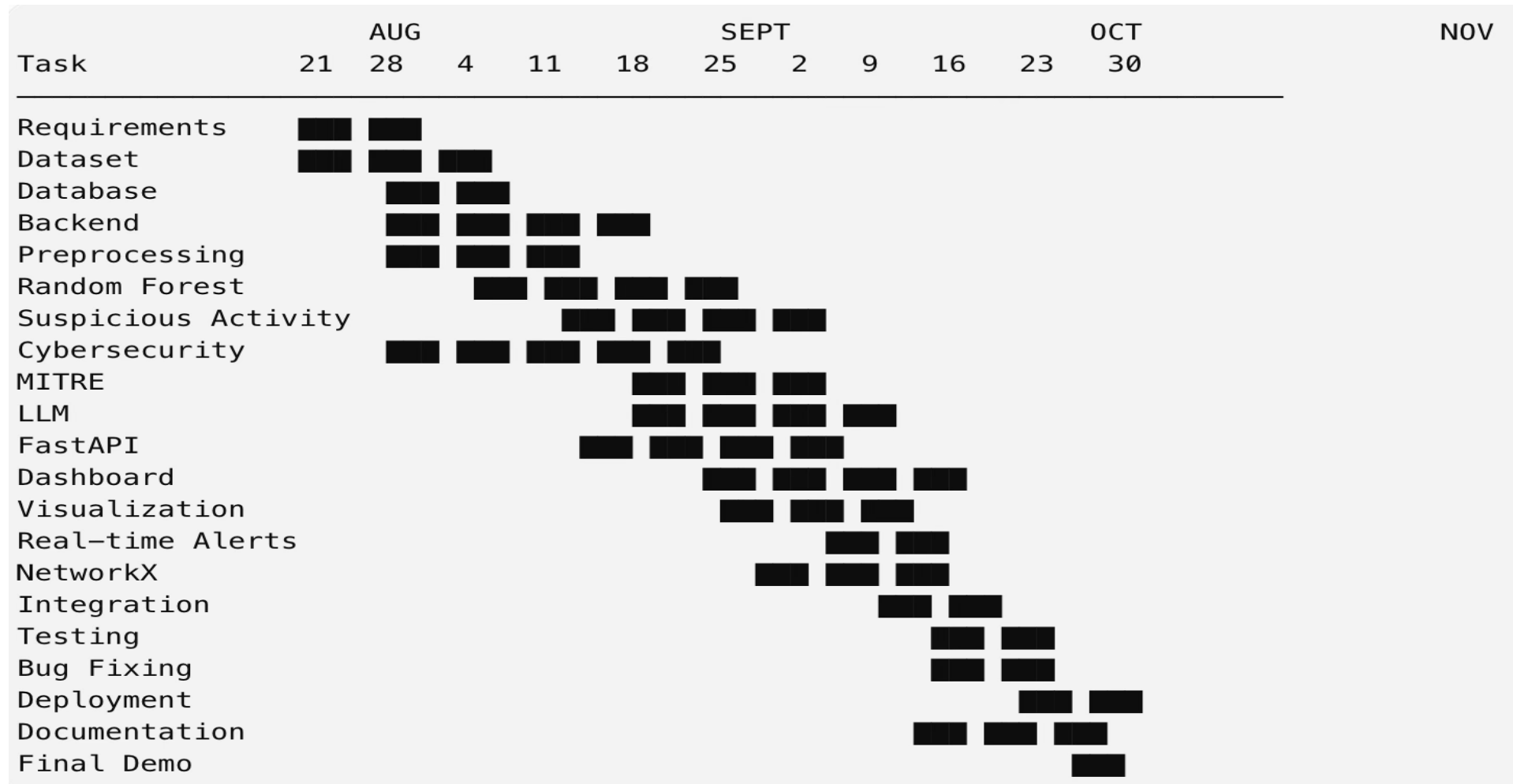
Literature Survey

Research Paper	Model(s) Used	Dataset	Key Results	FSIA Adaptation & Remarks
Toward Autonomous SOC Operations (2026)	Multi-Agent LLMs (GPT-4o, Llama 3.3)	SIEM logs (IBM QRadar, Google SecOps) and cloud security events	Achieved 82.8% threat detection accuracy with automated alert triage completed in under 10 minutes, reducing analyst workload and investigation time.	Provides the foundation for FSIA's investigation workflow. Instead of multiple large LLMs, FSIA uses a lightweight local LLM to reduce deployment cost while preserving automated threat investigation capabilities.
Agentic AI for Financial Crime Compliance (2025)	Agentic AI + LLM (n8n)	AML/FCC records, financial transactions, onboarding logs	Automated financial crime compliance monitoring, suspicious activity detection, and SAR generation, improving compliance efficiency while reducing manual effort.	FSIA extends the compliance framework beyond fraud detection by integrating cybersecurity incident investigation with automated, explainable reporting for regulatory compliance.
TITAN: Graph-Executable Reasoning (2025)	Path Planner + Graph Executor	TITAN dataset, MITRE ATT&CK	Demonstrated graph-based threat reasoning through relationship analysis, enabling accurate and explainable multi-step threat investigations.	FSIA adapts graph reasoning for live SOC environments by correlating multi-source security events into a unified attack timeline instead of only querying threat intelligence.
Incident Response Planning (2026)	DeepSeek-R1-14B + Lookahead Planning	68K incident-response pairs, CTU-Malware-2014, CIC-IDS-2017	Improved incident response with 22% faster recovery while reducing hallucinations in AI-generated remediation plans.	FSIA applies hallucination-reduction techniques throughout the investigation process—from threat detection and evidence collection to incident reporting and remediation recommendations.
Towards SLMs for Security Query Generation (2026)	Small Language Models (SLMs <10B)	NL2KQL dataset, Microsoft Sentinel logs	Achieved 98.7% query syntax accuracy with 10–15× lower computational cost, enabling efficient security log retrieval.	FSIA integrates lightweight query generation into its investigation pipeline, enabling automatic evidence collection with lower latency and operational cost than large cloud-based LLMs.

Redefined problem statement

Modern FinTech Security Operations Centers (SOCs) use AI for threat detection and alert prioritization, but analysts still spend considerable time collecting and correlating evidence from different security platforms, verifying AI-generated results, and preparing compliance reports. Existing solutions are often vendor-dependent, expensive due to cloud-based LLMs, and lack clear explanations for their decisions. This creates a need for a hybrid, explainable, and cost-effective AI framework that combines rule-based detection, anomaly analysis, and lightweight language models to automate evidence collection, incident investigation, and explainable security reporting.

Project Timeline



Project Review 1

Introduction

- The exponential growth of digital banking and online finance has increased vulnerability towards cyber attacks, fraud, and suspicious activity. FinTech companies are producing a massive amount of logs from transaction records, authentication methods, APIs, applications, cloud services, web application firewalls (WAFs), and firewalls. Traditional Security Operations Centers depend heavily on pre-set rules and manual investigation and thus cannot correlate events and deal with complex threats effectively.
- While LLMs can be helpful for security investigation purposes, applying them to all events can prove to be too costly and time-consuming since the volume of log data is too high and thus the number of tokens that should be used is increased as well. LLMs can sometimes hallucinate and give false results. Our project will help overcome these issues through utilizing machine learning to detect threats initially, LLMs selectively investigate and reason, relational graphs for correlating events, and explainable AI for auditing and cost-effective security analysis in FinTech environment.

Collect Security & Transaction Data

Identify Unusual Activity

Collect Evidence Automatically

Analyze and Recommend Actions

Generate Actionable Outputs

Clean and Prepare Data

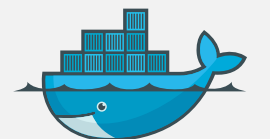
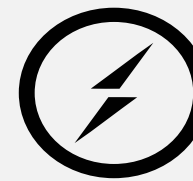
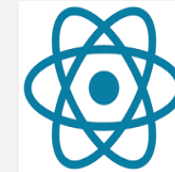
Flag High-Risk Activity

Connect Related Entities

Explain the Incident

Support Analyst Decisions

Tech Stack



Methodology

The proposed system follows a **hybrid AI-based security analysis approach** that combines machine learning, event correlation, knowledge graphs, and a lightweight language model. Security and transaction logs from authentication systems, APIs, WAFs, firewalls, cloud services, and applications are first **collected, cleaned, normalized, and converted into structured security features**. Machine learning techniques are then used to identify abnormal activities and assign risk scores to prioritize potentially malicious events.

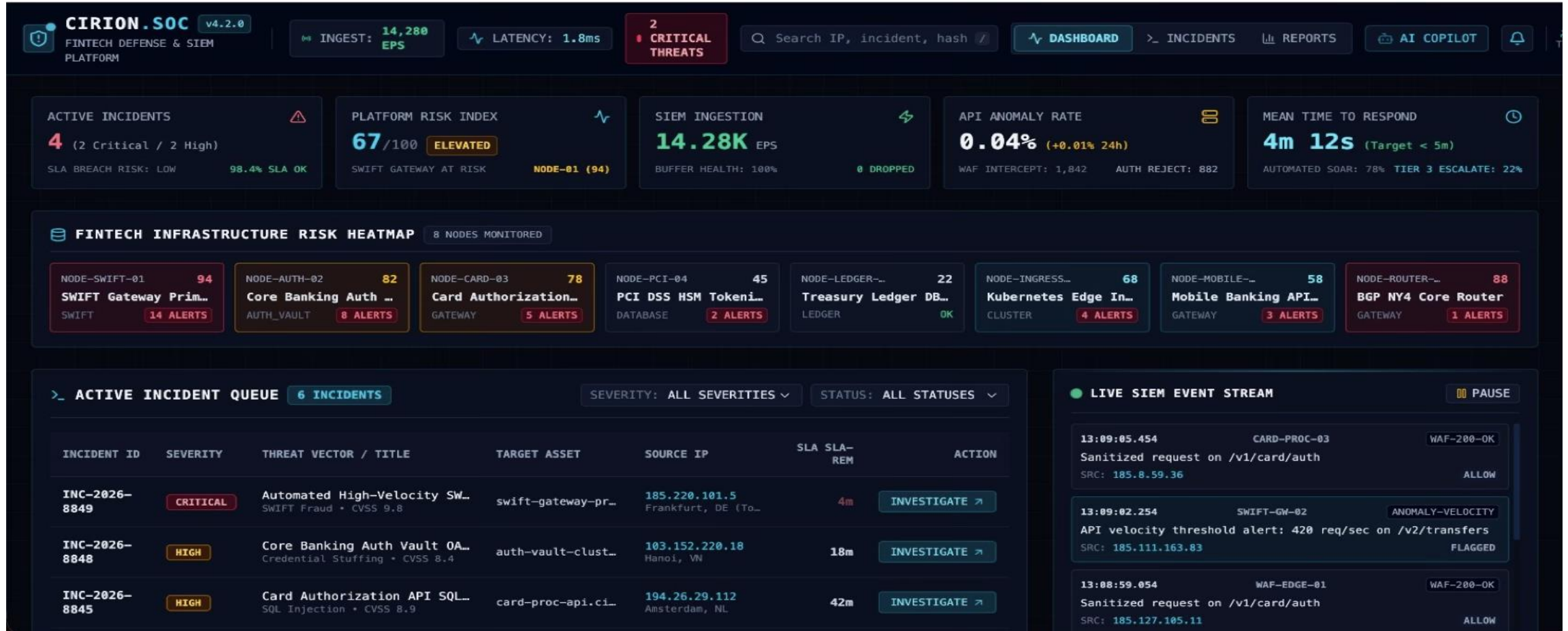
For high-risk activities, the system performs **multi-source event correlation** by linking related users, IP addresses, devices, transactions, and timestamps using a relational/knowledge graph. A lightweight LLM/SLM is selectively used to investigate these correlated incidents, retrieve relevant evidence, explain the possible attack pattern, and recommend appropriate actions. The final results are presented through an **explainable SOC dashboard**, including the incident timeline, risk level, supporting evidence, attack explanation, MITRE ATT&CK mapping, and recommended remediation actions.

Dataset description

Category	Attributes / Columns
Event Information	event_id , timestamp , log_source , event_type , severity
User & Session Information	user_id , session_id , device_id , source_ip , source_port , country
Authentication Information	authentication_status , login_method , failure_reason , mfa_status
API / HTTP Information	http_method , endpoint , status_code , response_time_ms , request_size_bytes , user_agent
Transaction Information	transaction_id , transaction_type , amount , currency , merchant_id , transaction_status
Network Information	destination_ip , destination_port , protocol , network_action , bytes_sent , bytes_received
Cloud / Application Information	service_name , resource_id , action , status , error_code
Security & Classification	attack_type , label , mitre_technique

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
user_id	session_id	device_id	source_ip	source_port	country	authentication_status	login_method	failure_reason	mfa_status	http_method	endpoint	status_code	response_time_ms	request_size_bytes	user_agent
U10747	S524415	DEV10921	10.42.41.92		IN										
U10156	S733120	DEV10247	10.134.25.64		IN					PUT	/api/v1/bee	200	24	748	Mozilla/
U10384	S253766	DEV10519	10.239.74.17		IN										
U10537	S613893	DEV11147	10.126.196.162		IN					PUT	/api/v1/proc	200	38	1990	Mozilla/
U10524	S894208	DEV10369	10.197.251.115		SG					POST	/api/v1/pa	200	33	865	Mozilla/
U10743	S136346	DEV10582	10.153.5.133		IN	SUCCESS	PASSWOR	NONE	NOT_REQUIRED						
U10784	S127170	DEV10342	10.151.87.	7966	IN										
U10488	S345674	DEV10328	10.197.25.133		IN										
U10544	S792711	DEV10948	10.92.61.6	20942	US										
U10339	S743966	DEV10412	10.207.34.88		IN										
U10475	S969659	DEV10592	10.168.249.43		SG	SUCCESS	BIOMETRIC	NONE	SUCCESS						
U10625	S286646	DEV10949	10.227.207.28		IN										
U10673	S668950	DEV10946	10.106.14.49		GB					GET	/api/v1/ba	200	31	786	Mozilla/
U10497	S330022	DEV10136	10.80.70.190		SG	SUCCESS	PASSWOR	NONE	SUCCESS						
U10684	S658254	DEV10235	10.239.22.	64232	IN										
U10243	S487371	DEV10508	10.229.16.68		US										
U10164	S746016	DEV10947	10.103.11.	56528	SG										
U10458	S819254	DEV10294	10.153.197.210		IN										
U10269	S952816	DEV10786	10.84.26.224		SG					PUT	/api/v1/bee	200	76	2366	Mozilla/
U10444	S736515	DEV10778	10.109.218.138		SG										
U10055	S469687	DEV10690	10.18.218.189		GB	SUCCESS	BIOMETRIC	NONE	SUCCESS						
U10082	S852896	DEV10945	10.43.121.	18188	GB										
U10210	S156644	DEV11017	10.130.220.193		IN					PUT	/api/v1/bee	200	73	1378	curl/8.1
U10142	S862469	DEV10391	10.21.107.12		IN	SUCCESS	PASSWOR	NONE	NOT_REQUIRED						
U10183	S879040	DEV10685	10.88.17.187		SG										
U10684	S549875	DEV10478	10.79.58.87		IN										
U10503	S632530	DEV11094	10.56.100.150		IN	FAILED	BIOMETRIC	UNKNOWN	FAILED						
U10113	S146266	DEV11136	10.16.197.	14604	IN										
U10459	S745594	DEV10836	10.105.26.240		GB										
U10076	S763976	DEV10374	10.142.228.126		IN										

Prototype of the frontend



Project Review 2 (5 marks)

Content

- Modifications as per suggestions, if any
- Results and discussion
(use multiple slides as per requirement)

Testing and results

Content

- Further Testing and results
- Modifications done if any

Project Review 3 (10 marks)

Content

- Results with tables\graphs\demonstration to prove achievement of objectives
- Conclusion with summary of findings from work done with justification

Report preparation and submission (Report typed on A-4 pages to be displayed)

Content

- Verification of report as per prescribed format
- Component\hardware bill submission, if any
- Return of temporary issued laboratory components\equipment if any